

1. Definition

Cloud computing is a technology that provides computing services like servers, storage, databases, and software over the internet.

Users can access these services without owning physical hardware.

It allows on-demand access to resources anytime and anywhere.

It helps reduce cost and improves flexibility.

2. Key Characteristics

- **It provides on-demand self-service to users.**
- **It allows access through internet from any device.**
- **It supports resource pooling (shared resources).**
- **It provides scalability to increase or decrease resources.**

3. Features

- **It follows pay-as-you-go pricing model.**
- **It provides high availability of services.**
- **It allows easy data storage and access.**

4. Advantages

- **It reduces infrastructure cost.**
- **It provides flexibility and scalability.**
- **It allows remote access to data and applications.**

5. Example

Google Drive or AWS cloud storage is used to store and access data online.

6. Conclusion

Cloud computing provides easy, scalable, and cost-effective IT services over the internet.

what is Virtualization

1. Definition

Virtualization is a technology that allows multiple virtual machines (VMs) to run on a single physical machine.

It creates a virtual version of hardware such as server, storage, or network.

Each virtual machine runs its own operating system independently.

It helps in efficient use of resources.

2. Types of Virtualization

- **Hardware virtualization allows multiple OS on one system.**
- **Storage virtualization combines multiple storage devices.**
- **Network virtualization creates virtual networks.**

3. Features

- **It allows multiple systems on one hardware.**
- **It improves resource utilization.**
- **It provides isolation between systems.**

4. Advantages

- **It reduces hardware cost.**
- **It improves system efficiency.**
- **It supports scalability.**

5. Example

Running Windows and Linux on the same computer using virtual machines.

1. Definition

Cloud computing works by providing services over the internet using remote servers.
Users access data and applications without storing them locally.
The cloud provider manages servers and resources in data centers.
It uses virtualization to deliver services efficiently.

2. Working Process

- **User sends request through internet.**
- **Request reaches cloud server.**
- **Server processes the request.**
- **Response is sent back to user.**

3. Components

- **Frontend is user device (mobile/computer).**
- **Backend includes servers and storage.**
- **Network connects user and cloud.**

4. Example

When a user uploads file on Google Drive, it is stored on cloud and accessed anytime.

->Types of Cloud

1. Definition

Types of cloud refer to different deployment models of cloud computing.
They define how cloud services are provided and managed.
Each type offers different level of security and control.
Organizations choose based on requirements.

2. Types of Cloud

->Public Cloud

Public cloud is available to general public over the internet.
Resources are shared among multiple users.
It is cost-effective and easy to use.
Example: AWS, Google Cloud.

->Private Cloud

Private cloud is used by a single organization only.
It provides high security and control over data.
Resources are not shared with others.
Example: Company's internal cloud system.

->Hybrid Cloud

Hybrid cloud is combination of public and private cloud.
It allows data to move between both environments.
It provides flexibility and scalability.
Example: Company stores sensitive data privately and uses public cloud for apps.

1. Definition

Cloud computing provides various benefits by delivering IT services over the internet.

It helps organizations reduce cost and improve efficiency.

It offers flexibility and scalability for applications.

It is widely used for modern computing needs.

2. Advantages

- **It reduces infrastructure and maintenance cost.**
- **It provides scalability to increase or decrease resources.**
- **It allows access from anywhere using internet.**
- **It provides data backup and recovery.**
- **It ensures high availability of services.**
- **It improves performance and speed.**

3. Example

A company stores data on cloud instead of buying physical servers.

AWS history

1. Definition

AWS (Amazon Web Services) is a cloud computing platform developed by Amazon.

It provides services like computing, storage, and networking over the internet.

It helps businesses build scalable and flexible applications.

It is one of the most popular cloud platforms in the world.

2. History

- **AWS was launched in 2006 by Amazon.**
- **First service introduced was Amazon S3 (Simple Storage Service).**
- **Later, EC2 (Elastic Compute Cloud) was introduced for virtual servers.**
- **AWS expanded with many services like database, networking, and analytics.**

3. Features

- **It provides global infrastructure.**
- **It supports scalable applications.**
- **It offers pay-as-you-go pricing model.**

4. Advantages

- **It reduces IT cost.**
- **It provides high availability.**

5. Example

Companies like Netflix use AWS for hosting applications.

1. Definition

AWS Dashboard is the main interface of AWS Management Console. It allows users to access and manage AWS services in one place. It provides an overview of resources and account information. It helps users easily navigate different AWS services.

2. Features

- **It displays list of all AWS services.**
- **It provides quick access to frequently used services.**
- **It shows account details and billing information.**
- **It allows monitoring of resources.**

3. Advantages

- **It is easy to use and user-friendly.**
- **It simplifies management of AWS resources.**

4. Example

User can launch EC2 instance or create S3 bucket using dashboard.

1. Definition

AWS (Amazon Web Services) is a cloud computing platform that provides IT services over the internet. It offers services like computing, storage, databases, and networking. It allows users to build scalable and flexible applications. It follows a pay-as-you-go pricing model.

2. Key Services

- **EC2 is used for virtual servers.**
- **S3 is used for storage.**
- **RDS is used for database services.**
- **VPC is used for networking.**

3. Features

- **It provides on-demand services.**
- **It supports global infrastructure.**
- **It offers high availability.**

4. Advantages

- **It reduces cost of infrastructure.**
- **It improves performance and scalability.**

5. Example

A company hosts its website on AWS using EC2 and stores data in S3.

1. Definition

Cloud architecture refers to the overall design and structure of cloud computing system. It defines how different components like servers, storage, and network work together. It includes both frontend (user side) and backend (cloud side). It ensures efficient and reliable delivery of cloud services.

2. Components of Cloud Architecture

- **Frontend is the user interface like browser or application.**
- **Backend includes servers, storage, database, and cloud services.**
- **Network connects frontend and backend through internet.**

3. Working

- **User sends request through frontend.**
- **Request goes to backend servers via network.**
- **Backend processes the request.**
- **Response is sent back to user.**

4. Features

- **It supports scalability of resources.**
- **It provides flexibility in system design.**
- **It ensures high availability.**

5. Advantages

- **It improves system performance.**
- **It ensures reliability and fault tolerance.**

6. Example

User accesses a cloud application through browser which connects to cloud servers.

Software as a Service (SaaS): Introduction

1. Definition

Software as a Service (SaaS) is a cloud computing model that provides software applications over the internet. Users can access these applications through a web browser without installing them. The service provider manages software, servers, and maintenance. It follows a subscription or pay-as-you-go model.

2. Key Features

- **Applications are accessed through internet.**
- **No need to install or maintain software.**
- **It supports multi-user access.**
- **Automatic updates are provided by provider.**

3. Working

- **User logs in through browser.**
- **Request goes to cloud server.**
- **Application runs on server.**
- **Output is displayed to user.**

4. Advantages

- **It reduces software cost.**
- **It provides easy access from anywhere.**
- **It eliminates maintenance work.**

5. Example : Google Docs and Gmail are examples of SaaS applications.

1. Definition : SaaS (Software as a Service) model provides software applications over the internet. Although it offers many benefits, it also has some limitations and challenges. These challenges affect security, performance, and control. Organizations must consider these issues before adopting SaaS.

2. Challenges in SaaS Model

- **Security Issues**
Data is stored on cloud, so there is risk of unauthorized access or data breach.
- **Data Privacy**
Sensitive data is handled by third-party providers, which may create privacy concerns.
- **Internet Dependency**
SaaS applications require continuous internet connection to work properly.
- **Limited Control**
Users have limited control over software and infrastructure.
- **Performance Issues**
Performance depends on internet speed and server availability.
- **Integration Problems**
It may be difficult to integrate SaaS with existing systems.
- **Downtime Risk**
Service may become unavailable if server fails.

3. Example : If internet connection is slow, users may face difficulty using online applications like Google Docs.

SaaS Integration Services

1. Definition

SaaS Integration Services are used to connect SaaS applications with other systems or applications. They help in sharing and exchanging data between different platforms. Integration ensures smooth communication between cloud and on-premise systems. It improves overall system efficiency and automation.

2. Need for Integration

- **Different applications need to work together.**
- **Data must be shared between systems.**
- **It reduces manual data entry.**

3. Types of Integration

- **Application Integration connects different software applications.**
- **Data Integration combines data from multiple sources.**
- **API Integration uses APIs to connect systems.**

4. Features

- **It enables real-time data exchange.**
- **It supports automation of processes.**
- **It improves system connectivity.**

5. Advantages

- **It saves time and effort.**
- **It reduces errors.**
- **It improves productivity.**

6. Example

Integrating CRM software with email system to automatically update customer data.

1. Definition

SaaS (Software as a Service) provides software applications over the internet.

Users can access applications without installing them on local devices.

It is managed by the service provider.

It offers both benefits and limitations.

2. Advantages of SaaS

- **It reduces software and hardware cost.**
- **It allows access from anywhere using internet.**
- **It does not require installation or maintenance.**
- **It provides automatic updates.**
- **It supports scalability for users.**

3. Disadvantages of SaaS

- **It depends on internet connection.**
- **It may have security and privacy issues.**
- **It provides limited control over software.**
- **Performance depends on network speed.**
- **Integration with other systems may be difficult.**

Infrastructure As a Services(IaaS): Introduction

1. Definition

Infrastructure as a Service (IaaS) is a cloud computing model that provides virtualized computing resources over the internet.

It includes services like virtual machines, storage, and networking.

Users do not need to manage physical hardware.

It provides flexibility and scalability for infrastructure.

2. Key Components

- **Virtual machines (servers)**
- **Storage (disk, backup)**
- **Networking (IP, firewall)**

3. Features

- **It provides on-demand infrastructure.**
- **It allows users to scale resources easily.**
- **It follows pay-as-you-go model.**

4. Advantages

- **It reduces hardware cost.**
- **It provides flexibility and control.**
- **It supports fast deployment.**

5. Example

Using AWS EC2 to create virtual servers instead of physical servers.

1. Definition

Virtual Machine (VM) is a software-based computer that runs on a physical machine. It behaves like a real computer with its own operating system and applications. Multiple virtual machines can run on a single physical system. It is created using virtualization technology.

2. Features

- **VM runs its own operating system independently.**
- **Multiple VMs can run on one physical machine.**
- **It provides isolation between systems.**

3. Advantages

- **It improves resource utilization.**
- **It reduces hardware cost.**
- **It provides flexibility and scalability.**

4. Example

Running Windows and Linux as separate virtual machines on the same computer.

VM Migration Services

1. Definition

VM Migration Services are used to move virtual machines from one physical server to another. It helps in balancing load and maintaining system performance. Migration can be done without stopping the VM (live migration). It is important for cloud management and maintenance.

2. Types of Migration

- **Live Migration moves VM without shutting it down.**
- **Cold Migration moves VM after stopping it.**

3. Features

- **It allows movement of VMs between servers.**
- **It supports load balancing.**
- **It reduces downtime.**

4. Advantages

- **It improves system performance.**
- **It ensures high availability.**
- **It helps in maintenance without interruption.**

5. Example

Moving a VM from overloaded server to another server to balance load.

Advantages and Disadvantages of IaaS

1. Definition

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. It includes servers, storage, and networking services. Users can manage infrastructure without owning physical hardware. It offers both benefits and limitations.

2. Advantages of IaaS

- It reduces cost of physical hardware.
 - It provides high scalability of resources.
 - It allows full control over infrastructure.
 - It supports fast deployment of servers.
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3. Disadvantages of IaaS

- It requires technical knowledge to manage resources.
 - It may have security risks if not configured properly.
 - It depends on internet connection.
 - It may lead to higher cost if not managed properly.
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Platform as a Service (PaaS): Introduction

1. Definition

Platform as a Service (PaaS) is a cloud computing model that provides a complete platform for developing, testing, and deploying applications over the internet. It includes development tools, libraries, and runtime environment. Users do not need to manage hardware or operating system. It helps developers focus only on application development.

2. Key Components

- Development tools and frameworks are provided.
- Database management system is included.
- Runtime environment is available for execution.

3. Features

- It provides ready-to-use development platform.
- It supports application deployment and testing.
- It offers scalability for applications.

4. Advantages

- It reduces development time.
- It eliminates hardware management.
- It supports collaboration among developers.

5. Example

Google App Engine and Microsoft Azure App Services are examples of PaaS.

1. Definition

Integration of Private Cloud refers to connecting a private cloud with other systems or cloud environments. It allows data and applications to be shared between private cloud and external services. It improves flexibility and resource utilization. It is commonly used in hybrid cloud environments.

2. Need for Integration

- **Organizations need to connect private cloud with public cloud.**
- **It helps in sharing data and applications.**
- **It improves system performance and scalability.**

3. Methods of Integration

- **API Integration is used to connect applications.**
- **Network Integration connects systems using secure networks.**
- **Data Integration helps in sharing and synchronizing data.**

4. Features

- **It enables smooth communication between systems.**
- **It supports real-time data exchange.**
- **It improves flexibility of cloud usage.**

5. Advantages

- **It provides better resource utilization.**
- **It improves scalability.**
- **It ensures secure data transfer.**

6. Example

A company uses private cloud for sensitive data and integrates it with public cloud for applications.

Public Cloud

1. Definition

Public Cloud is a type of cloud computing where services are provided over the internet to multiple users. The cloud infrastructure is owned and managed by a third-party provider. Resources like servers and storage are shared among users. It is easily accessible and cost-effective.

2. Features

- **It is available to general public over internet.**
- **Resources are shared among multiple users.**
- **It provides high scalability.**
- **It follows pay-as-you-go model.**

3. Advantages

- **It reduces cost of infrastructure.**
- **It provides easy access from anywhere.**
- **It supports scalability and flexibility.**

4. Disadvantages

- **It has lower security compared to private cloud.**
- **Resources are shared with other users.**
- **It depends on internet connection.**

5. Example : AWS, Microsoft Azure, and Google Cloud are examples of public cloud.

Advantages and Disadvantages of PaaS

1. Definition

Platform as a Service (PaaS) provides a platform for developing, testing, and deploying applications over the internet.

It includes tools, frameworks, and runtime environment.

Users do not manage hardware or operating system.

It offers both benefits and limitations.

2. Advantages of PaaS

- **It reduces application development time.**
 - **It eliminates need to manage hardware and OS.**
 - **It supports easy deployment of applications.**
 - **It provides scalability for applications.**
 - **It allows collaboration among developers.**
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3. Disadvantages of PaaS

- **It provides limited control over infrastructure.**
- **It may create dependency on service provider (vendor lock-in).**
- **Security may be a concern for sensitive data.**
- **It may have compatibility issues with some applications.**